DS UA 9201 Problem Set 1

This homework must be turned in on Brightspace by **Oct 11th 2021, 11:59pm (Paris time)**. It must be your own work, and your own work only—you must not copy anyone's work, or allow anyone to copy yours. This extends to writing code. You may consult with others, but when you write up, you must do so alone. Your homework submission must be written and submitted using Rmarkdown. No handwritten solutions will be accepted. You should submit:

- A compiled PDF file named `yourNetID_solutions.pdf` containing your solutions to the problems.
- A `.ipynb` file containing the code and text used to produce your compiled pdf named `yourNetID_solutions.ipynb`. Note that math can be typeset in markdown cells (in the jupyter notebook) in the same way as Latex.

Please make sure your answers are clearly structured in the jupyter notebook file:

- Label each question (e.g. 1.1 for exercise 1, question 1).
- Do not include written answers as code comments.
- The code used to obtain the answer for each question part should accompany the written answer.

Exercise 1: Salk Vaccine Field Trial (15 points)

Polio claimed hundreds of thousands of victims from 1916-1956. It affected mainly children. By the 1950s, several vaccines had been discovered and successfully tested in the lab. The most promising one had been proposed by Jonas Salk.
By 1954 public health service was ready to try the vaccine in the real word, i.e., outside the lab on patients.

Question 1 (5 points)

What if the vaccine had just be given to a large number of children in 1954? Would this have allowed to conclude on the efficiency of the treatment, for instance by looking at the polio cases in the following year?

Question 2 (5 points)

Instead, the vaccine is given to a treatment group and not to a control group. Treatment and control group are defined via the parent's permission to treat their children. (parent's permission is required to vaccinate children). Is this a valid randomization strategy to test the effect of the vaccine? Justify your answer.

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Question 3 (5 points)

What is a better alternative to the above strategy?

Exercise 2: Average Treatment Effect on the Treated (25 points)

Suppose we have a sample of observations, each assigned a binary treatment $T_i \in \{0,1\}$ with $T_i = 1$ indicating a unit is treated and $T_i = 0$ indicating the unit is assigned control. Assume $0 < Pr(T_i = 1) < 1$. We observe an outcome Y_i for each observation. We define potential outcomes $Y_i(1)$ and $Y_i(0)$, denoting the outcome observed for unit i if it were assigned treatment ($Y_i(1)$) or control ($Y_i(0)$) respectively. Assume that $Y_i(1), Y_i(0)$ are iid from the same distribution, which implies that: $E[Y_1(t)] = \cdots = E[Y_n(t)] = \mu(t)$ and $Var[Y_1(t)] = \cdots = Var[Y_n(t)] = \sigma^2(t)$.

In class we have talked about the Average Treatment Effect (ATE), which is defined as $\tau = E\left[Y_i(1) - Y_i(0)\right]$. Another very common effect of interest to researchers is the Average Treatment Effect on the Treated (ATT), which is defined as: $\tau^t = E[Y_i(1) - Y_i(0)|T_i = 1]$.

Question 1 (5 points)

What is the interpretation of the ATT? Give your description of what this effect means and how it is different from the ATE.

Question 2 (10 points)

We assume that:

- The Stable Unit Treatment Values Assumption holds for all treatment levels: $Y_i = Y_i(1)T_i + Y_i(0)(1 - T_i)$
- Weak ignorability holds only for the control outcome, i.e.: $Y_i(0) \perp\!\!\!\perp T_i$, and it is not true that: $Y_i(1) \perp\!\!\!\perp T_i$

Show that consistency of all treatment levels, and weak ignorability of the control contition (the assumptions just made) are enough to identify the ATT, i.e., show that: $\tau^t = E[Y_i|T_i = 1] - E[Y_i|T_i = 0]$.

Question 3 (10 points)

Write and simplify the difference between the ATE and the ATT under the same assumptions as the previous question. What additional assumption is necessary for this difference to be 0, and for the ATT to be equal to the ATE? Why is this assumption enough?

indication: You may start from the definition $\tau - \tau^t = E[Y_i(1) - Y_i(0)] - E[Y_i(1) - Y_i(0)|T_i = 1]$

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Exercise 3: Bernoulli trial (30 points)

Under the same setting as Exercise 2, we suppose that T_i is assigned to the n units in a Bernoulli trial, that is each unit receives treatment independently with probability $Pr(T_i = 1) = p$.

Question 1 (10 points)

Recall that the number of treated units, N_t is defined as $N_t = \sum_{i=1}^{n} T_i$. Recall that in the case of a Bernoulli trial, N_t is a random variable.

1. What distribution does T_i follow?
2. What is $E[N_t]$?
3. What is $Var[N_t]$?
4. Suppose that we wanted the expected number of treated units in our Bernoulli trial to be the same as the number of treated units as a completely randomized experiment with n_t treated units. What value of p should we choose?

Question 2 (15 points)

After conducting the experiment as described above, we wish to estimate the ATE. To do so, we employ the following estimator:

$$\hat{\tau}_{IPW} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i\frac{T_i}{p} - Y_i\frac{1-T_i}{1-p}\right).$$

Show that under consistency, positivity, and ignorability for all treatments this estimator is unbiased for the ATE.

Question 3 (5 points)

Suppose now that we used the same estimator defined above in a **completely randomized experiment**, where exactly n_t units are treated. Show that, in this case, the estimator above is equal to the Neyman "difference-in-means" estimator we saw in class.

Exercise 4: Social Pressure and Voter Turnout (55 points)

Gerber, Green and Larimer randomly assigned households to receive a mailing encouraging them to turn out to vote before the Michigan 2006 primary election (Gerber, Green and Larimer (APSR, 2008)). We will be using the individual data obtained

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from the experiment. Each row in the dataset represent an individual record, where `p2000` represents whether the individual had voted in August 2000, `g2000` represents whether the individual had voted in November 2000 (same for `p2002`, `g2002`, `p2004`). Each individual belongs to a household specified by `hh_id`.

Question 1 Data preparation (10 points)

In order to analyze the GOTV data we will need to reproduce the household-level dataset of the original paper.

1. Recode the variable `sex` by changing the character to float (i.e. "female" to 1., "male" to 0.)
2. Recode the variable `yob` (year of birth) into a new variable called "age" by subtracting `yob` from the year the experiment took place, 2006.
3. Group the data into households, i.e., create a new dataframe where each row is a household with a unique `hh_id`, and each column is the the mean value of each of the other individual-level variables in that household. (Hint: you may consider using the `groupby` method in `pandas`.)
4. In the paper, the authors analyzed households rather than individual. Why did they do this?

Question 2 Validate Randomization (10 points)

Using the household dataset you obtained above, verify that the experimental assignment is randomized at the household level by computing and showing the sample means of each of the variables: `p2000`, `g2000`, `p2002`, `g2002`, `p2004`, `hh_id`, `sex`, and `age` in each of the treatment groups. Are these means similar across groups? And if so what does that imply for randomization and ignorability? Is it enough to prove that the treatment was indeed randomized if we don't know it from the experimenters?

Question 3 Average Treatment Effect (5 points)

Use the household dataset you obtained above, use the Neyman Estimator, denoted here as $\hat{\tau}$, to compute the average treatment effect for each treatment group comparing to the control group. Name and briefly explain two assumptions in this experiment that allow us to compute the ATE.

Question 4 Variance and hypothesis testing (10 points)

Assuming that the experiment is a completely randomized experiment, give an estimate of the ATE variance of the treatment effect of the **Neighbors** treatment compared to the control group, using the Neyman variance estimator, denoted as $\widehat{Var}[\hat{\tau}]$. In addition, conduct a two-sided hypothesis test against the null that the

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ATE is 0, i.e.: $H0 : \tau = 0$, with the alternative is $H1 : \tau \neq 0$, using the Z-statistic as your test statistic, i.e.:

$$Z_n = \frac{\sqrt{n}(\hat{\tau} - \tau)}{\sqrt{\widehat{Var}[\hat{\tau}]}}.$$

Report both the value of Z_n and the p-value for the test.

Question 5 Randomization Inference (15 points)

Conduct a randomization inference hypothesis test on the experiment data for the sharp null hypothesis that $Y_i(\text{neighbors}) = Y_i(\text{control})$ for all i. Using Z_n as defined before as your test statistic, follow the steps below:

1. Simulate the value of Z_n under the sharp null for at least $N = 1000$ iterations.
2. Plot the values you obtained as a histogram.
3. Add a marker for the observed value of Z_n.
4. Report the two-sided p-value for the test.

Question 5.1 Compare hypothesis tests (5 points)

Briefly comment on the difference between the p-value you obtained using those two different strategies. Which is smaller? And what could this difference be due to?